

AIO 2026 COMPETITION SUBMISSION

CardioInsight

AI-Powered ECG Diagnosis System with Explainable
Artificial Intelligence and LLM Report Generation

CardioInsight combines a hybrid CNN-SE-Transformer deep learning model, three complementary explainability methods (Grad-CAM, Saliency Maps, Integrated Gradients), and a fine-tuned GPT-4o-mini report generator into an end-to-end system for automated 12-lead ECG classification and clinical interpretation.

Category: Eco Innovation / Machine Learning

Dataset: PTB-XL (21,837 ECG recordings)

AI Methods: CNN, Transformer, XAI, LLM

Project Team: Ibrahim Melhem
Abdallah Jbr

Team Name: bau-AI

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Chapter 1: Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization. The electrocardiogram (ECG) is the most widely used non-invasive diagnostic tool for detecting cardiac abnormalities, yet its interpretation demands specialized expertise that is often unavailable in resource-constrained settings. Misinterpretation or delayed diagnosis of ECG recordings can lead to adverse clinical outcomes, particularly in emergency scenarios where rapid triage decisions are critical. This gap between the ubiquity of ECG recordings and the scarcity of expert interpreters constitutes a significant public health challenge, one that is especially acute in regions with limited access to trained cardiologists.

CardioInsight addresses this challenge by developing an AI-powered ECG analysis system that combines deep learning classification with explainable artificial intelligence (XAI) and large language model (LLM) report generation. The system classifies 12-lead ECG recordings into five diagnostic categories: Normal Sinus Rhythm (NORM), Myocardial Infarction (MI), ST/T-Wave Change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). Beyond classification, CardioInsight provides three complementary explainability methods (Grad-CAM, Saliency Maps, and Integrated Gradients) that reveal which signal regions and patient metadata features drive each prediction, and generates structured clinical interpretation reports through a fine-tuned GPT-4o-mini model.

The project targets the Eco Innovation theme of the Arab Artificial Intelligence Olympiad (AIO) 2026, aligning with the goal of harnessing AI to create smart, sustainable solutions that protect health and improve lives. By democratizing access to expert-level ECG interpretation, CardioInsight has the potential to reduce diagnostic delays, support clinical decision-making in underserved areas, and ultimately contribute to better cardiovascular health outcomes on a global scale.

1.1 Objectives

The primary objective of this project is to design, implement, and validate an end-to-end AI system for automated ECG diagnosis that is both clinically useful and technically transparent. The specific aims are:

- To develop a hybrid CNN-SE-Transformer deep learning model capable of multi-label ECG classification across five diagnostic categories with per-class F1-optimized thresholds.
- To implement three XAI methods (Grad-CAM, Saliency Maps, and Integrated Gradients) that provide lead-level and metadata-level attribution, making model predictions interpretable to clinicians.

- To integrate a fine-tuned GPT-4o-mini model for automated generation of structured clinical interpretation reports from model predictions and patient metadata.
- To build a real-time streaming pipeline with a binary pre-filter that enables continuous ECG monitoring with efficient computation.
- To deliver a functional Streamlit-based web application supporting manual upload, batch processing, and live monitoring modes.

1.2 Scope

This project focuses on the analysis of standard 12-lead ECG recordings in the WFDB format, sourced from the PTB-XL dataset, the largest publicly available clinical ECG dataset. The system processes raw signal data through a multi-stage pipeline encompassing signal preprocessing (bandpass filtering and wavelet denoising), feature extraction through deep learning, multi-label classification, explainability analysis, and automated report generation. The application supports three operational modes: single-record manual upload, batch processing of multiple records, and real-time streaming analysis. While the system demonstrates strong classification performance, it is designed as a clinical decision support tool and not as a replacement for professional medical diagnosis. A research-use-only disclaimer is prominently displayed in the application interface.

Chapter 2: Background and Literature Review

2.1 ECG Fundamentals and Diagnostic Categories

The electrocardiogram records the electrical activity of the heart through electrodes placed on the body surface, producing a time-series signal across multiple leads. A standard 12-lead ECG provides twelve simultaneous views of the cardiac electrical cycle, each capturing a different spatial perspective on the heart. The P wave represents atrial depolarization, the QRS complex corresponds to ventricular depolarization, and the T wave reflects ventricular repolarization. Abnormalities in the morphology, timing, or amplitude of these features are indicative of specific cardiac conditions. The five diagnostic superclasses addressed by CardioInsight encompass the most prevalent ECG abnormalities: Normal Sinus Rhythm (NORM) indicates healthy cardiac function; Myocardial Infarction (MI) signals heart muscle damage due to coronary artery occlusion; ST/T-Wave Change (STTC) encompasses repolarization abnormalities; Conduction Disturbance (CD) covers defects in the electrical conduction system; and Hypertrophy (HYP) indicates abnormal thickening of the heart muscle.

2.2 Deep Learning for ECG Analysis

Traditional ECG interpretation relies on rule-based algorithms and manual review by cardiologists. However, the advent of deep learning has enabled automated ECG analysis with performance approaching or exceeding that of human experts in specific tasks. Convolutional Neural Networks (CNNs) have been widely adopted for ECG classification due to their ability to extract local temporal features through convolutional filters. Hannun et al. (2019) demonstrated that a deep CNN could classify 12 rhythm classes with cardiologist-level accuracy. More recently, Transformer-based architectures have shown promise for ECG analysis by capturing long-range temporal dependencies through self-attention mechanisms. The SE (Squeeze-and-Excitation) block, introduced by Hu et al. (2018), provides channel-wise attention that adaptively recalibrates feature maps, and has been shown to improve ECG classification performance when integrated into CNN backbones. The PTB-XL dataset, published by Wagner et al. (2020), is the largest publicly available clinical 12-lead ECG dataset with 21,837 records from 18,885 patients, annotated with 71 diagnostic statements organized into a hierarchical superclass system. This dataset has become the de facto benchmark for ECG classification research.

2.3 Explainable AI in Medical Diagnostics

The black-box nature of deep learning models poses a significant barrier to clinical adoption, as physicians require transparency in diagnostic reasoning. Explainable AI (XAI) methods aim to bridge this gap by providing insights into model decision-making processes. Grad-CAM (Gradient-weighted Class Activation Mapping), proposed by Selvaraju et al. (2017), generates visual heatmaps highlighting the input regions most influential to a specific prediction by computing gradients of the target class with respect to the final convolutional layer. Saliency Maps compute the gradient of the output with respect to the input, providing a pixel-level attribution map. Integrated Gradients (IG), introduced by Sundararajan et al. (2017), offers an axiomatic approach by accumulating gradients along a straight-line path from a baseline input to the actual input, satisfying both the sensitivity and implementation invariance axioms. In the medical domain, these methods have been applied to ECG interpretation to identify diagnostically relevant signal segments, providing clinicians with visual evidence supporting or questioning model predictions. The integration of multiple complementary XAI methods in CardioInsight enables a more robust and multi-faceted understanding of model behavior.

2.4 LLM-Assisted Clinical Reporting

Large language models (LLMs) have demonstrated significant capability in generating fluent, contextually appropriate text from structured inputs. In the medical domain, fine-tuned LLMs have been explored for automated report generation from imaging and signal data. By conditioning on model predictions and patient metadata, an LLM can synthesize a coherent clinical interpretation that integrates diagnostic findings with patient context. This approach mirrors the structure of professional ECG reports, which typically include a diagnostic summary, clinical correlation,

possible symptoms, and recommendations. CardioInsight leverages a fine-tuned GPT-4o-mini model with a carefully engineered prompt template that encodes diagnostic predictions, confidence scores, heart rate, rhythm assessment, and patient demographics to generate structured medical reports. The use of streaming inference ensures responsive report delivery, while retry logic with exponential backoff provides resilience against transient API failures.

2.5 Existing Solutions and Gaps

Several commercial and research systems have been developed for automated ECG interpretation, including the Glasgow Algorithm, the Marquette 12SL system, and more recently, AI-based solutions from companies such as AliveCor and Cardiologs. However, existing solutions exhibit notable limitations: most provide only classification outputs without explainability, making it difficult for clinicians to trust or verify predictions; few integrate LLM-powered report generation, requiring manual interpretation and documentation; and real-time streaming analysis with efficient pre-filtering remains uncommon. CardioInsight addresses these gaps by combining multi-label classification, three complementary XAI methods, LLM report generation, and a streaming pipeline with binary pre-filtering into a single, cohesive system.

Chapter 3: Solution Design

3.1 System Architecture Overview

CardioInsight follows a modular, pipeline-based architecture composed of six core modules: (1) the Signal Processor, which handles ECG preprocessing; (2) the Metadata Processor, which normalizes patient demographic data; (3) the ECG Engine, which manages model loading and inference; (4) the XAI Engine, which provides explainability analysis; (5) the GPT Report Generator, which produces clinical reports; and (6) the API Layer, which orchestrates the full diagnostic pipeline. The Streamlit frontend provides three operational modes: manual upload, batch processing, and live monitoring. All components communicate through well-defined data contracts, enabling independent testing and future extensibility.

Module	File	Responsibility
Signal Processor	Signal_Processor.py	Bandpass filtering, wavelet denoising, z-score normalization, fixed-length resampling
Metadata Processor	Metadata_Processor.py	Z-score scaling of age/height/weight, sex encoding, feature vector assembly
ECG Engine	ecg_engine.py	Model loading (.keras and config.json+H5), binary pre-filter, heart rate/rhythm estimation

XAI Engine	xai_engine.py	Grad-CAM, Saliency Maps, Integrated Gradients with adaptive lead visualization
GPT Report	gpt_report.py	Fine-tuned GPT-4o-mini report generation with streaming and retry logic
API Pipeline	api_bassam.py	Full pipeline orchestration, inference, thresholding, streaming windowed analysis
Frontend	Copy2App.py	Streamlit UI with manual, batch, and live monitoring modes

Table 1: System module architecture and responsibilities

3.2 Signal Preprocessing Pipeline

The signal preprocessing pipeline transforms raw ECG recordings into model-ready inputs through a sequence of noise reduction and normalization steps. First, the raw signal is loaded from WFDB-format .hea/.dat file pairs using the wfdb library, which provides the signal matrix and sampling frequency. A 4th-order Butterworth bandpass filter with cutoff frequencies of 0.5 Hz and 45 Hz removes baseline wander and high-frequency noise while preserving the diagnostic frequency content of the ECG. Subsequently, wavelet-based denoising using the Daubechies-4 (db4) wavelet at decomposition level 2 applies soft thresholding to detail coefficients, with the universal threshold scaled by 0.5 to preserve subtle diagnostic features. Each lead is then z-score normalized using pre-computed mean and standard deviation values derived from the PTB-XL training set. Finally, the signal is truncated or zero-padded to a fixed length of 1000 samples and reshaped to the (1, 1000, 12) format expected by the model.

3.3 Hybrid CNN-SE-Transformer Model Architecture

The classification model employs a hybrid architecture that integrates convolutional feature extraction, channel attention, and temporal self-attention. The signal branch begins with 1D convolutional layers that extract local temporal features from the preprocessed ECG signal. An SE (Squeeze-and-Excitation) block is applied after the convolutional layers, performing global average pooling followed by two fully connected layers to compute channel-wise attention weights. These weights adaptively recalibrate the feature maps, amplifying diagnostically relevant channels while suppressing noise. The attended features are then processed by a Transformer Encoder, which applies multi-head self-attention to capture long-range temporal dependencies across the signal. The Transformer block includes layer normalization, a feed-forward network with dropout regularization, and residual connections. The metadata branch processes patient demographic information (age, sex, height, weight) through a separate dense pathway. The signal and metadata representations are fused and passed through dense layers with sigmoid activation for multi-label classification. The model is trained using focal loss ($\gamma=2.0$, $\alpha=0.25$) with label smoothing

(epsilon=0.1) to address class imbalance and improve generalization on rare conditions.

3.4 Binary Pre-Filter for Streaming Efficiency

Real-time ECG monitoring requires processing continuous signal streams efficiently. CardioInsight implements a two-stage inference architecture for streaming mode. A lightweight binary classifier first evaluates each signal window, producing a normal/abnormal prediction with a confidence score. Only windows flagged as abnormal by the binary pre-filter are forwarded to the full multi-label model for detailed diagnosis. This architecture significantly reduces computational overhead in real-time scenarios where the majority of windows may represent normal sinus rhythm. The binary model is loaded from a config.json + model.h5 pair and operates on the preprocessed signal alone, without requiring patient metadata, enabling rapid triage decisions. If the binary model is unavailable, the system gracefully degrades by routing all windows to the full model, ensuring continuous operation.

3.5 Multi-Label Classification with Optimized Thresholds

ECG diagnosis frequently involves multiple co-occurring conditions, necessitating multi-label rather than multi-class classification. CardioInsight applies per-class F1-optimized thresholds to convert sigmoid probabilities into binary predictions. These thresholds (NORM: 0.61, MI: 0.49, STTC: 0.44, CD: 0.35, HYP: 0.42) are derived from validation set optimization and reflect the varying base rates and clinical importance of each condition. A post-processing rule resolves the logical conflict that arises when both NORM and one or more pathological conditions exceed their thresholds: in such cases, the NORM label is suppressed, as the presence of a pathological condition takes clinical precedence. The prediction with the highest probability among the detected conditions is identified as the primary diagnosis.

Class	Full Name	Thresho Id	Description
NOR M	Normal Sinus Rhythm	0.61	Healthy cardiac function
MI	Myocardial Infarction	0.49	Heart muscle damage from coronary occlusion
STTC	ST/T-Wave Change	0.44	Repolarization abnormalities
CD	Conduction Disturbance	0.35	Electrical conduction system defects
HYP	Hypertrophy	0.42	Abnormal heart muscle thickening

Table 2: Per-class F1-optimized classification thresholds

3.6 Explainability Integration

CardioInsight implements three complementary XAI methods, each providing a distinct perspective on model predictions. Grad-CAM generates class-discriminative heatmaps by computing gradients of the target class score with respect to the output of the last convolutional layer. The resulting class activation map is upsampled and overlaid on the raw ECG waveform, with per-lead signal modulation to produce lead-specific heatmaps that highlight diagnostically relevant temporal regions. A robust layer-finding algorithm with multiple fallback strategies ensures compatibility with Keras 3.x functional API models. Saliency Maps compute the gradient of the target class output with respect to the input signal, producing a pixel-level attribution map. Regions with saliency values exceeding 0.3 are highlighted as areas of high influence. Integrated Gradients (IG) accumulates gradients along an interpolation path from a zero baseline to the actual input across 50 steps, batched for efficiency. IG provides both signal-level and metadata-level attributions: the mean absolute IG signal values across time yield per-lead importance scores, while the IG metadata values indicate the influence of each demographic feature. These attributions are visualized through bar charts for lead importance and metadata impact, and through overlay plots for signal-level attribution.

3.7 LLM Report Generation

The report generation module uses a fine-tuned GPT-4o-mini model accessed through the OpenAI API. A system prompt defines the model as an expert cardiology AI assistant specialized in ECG interpretation, specifying the required report structure: diagnostic summary, clinical correlation, possible symptoms, and recommendations. The user prompt encodes detected conditions with their confidence scores, patient demographics (age, sex), heart rate, rhythm assessment, and regularity. Streaming inference is employed to deliver the report incrementally, improving perceived responsiveness. A three-attempt retry mechanism with exponential backoff (minimum 2 seconds, maximum 10 seconds) ensures resilience against transient API failures. When the LLM is unavailable, a structured fallback report is generated from the prediction data and patient metadata, ensuring that the system always provides a clinical summary.

3.8 Streaming and Batch Modes

The streaming pipeline processes continuous ECG signals by slicing them into overlapping 10-second windows with 5-second overlap. Each window undergoes the full preprocessing pipeline and is first evaluated by the binary pre-filter. Abnormal windows trigger the full multi-label model for detailed diagnosis, while normal windows return a clean NORM prediction with the binary confidence score. The streaming generator yields per-window results including predictions, heart rate, rhythm assessment, anomaly flags, and timing information, enabling real-time visualization in

the Streamlit interface. The batch pipeline processes multiple ECG records sequentially, supporting file-pair and ZIP archive uploads. Batch results are presented as a summary table with downloadable CSV output, enabling efficient screening of large ECG datasets.

Chapter 4: Data Analysis and Results

4.1 Dataset Description

The model was trained and evaluated on the PTB-XL dataset, the largest publicly available clinical 12-lead ECG dataset. PTB-XL contains 21,837 recordings from 18,885 patients, sampled at 500 Hz with durations ranging from 10 seconds to 60 seconds. Each recording is annotated with up to 71 diagnostic statements organized into a five-class superclass hierarchy: NORM, MI, STTC, CD, and HYP. The dataset exhibits significant class imbalance, with NORM comprising approximately 42% of all records, while rare conditions such as HYP account for less than 10%. Patient demographics include age (mean 59.6 years, standard deviation 16.9), sex (approximately 61% male), height (mean 167.5 cm), and weight (mean 71.7 kg). The dataset was split into training (approximately 80%) and test (approximately 20%) sets following the recommended PTB-XL fold assignments, ensuring no patient overlap between splits.

Attribute	Value
Total recordings	21,837
Unique patients	18,885
Sampling frequency	500 Hz
Duration range	10 - 60 seconds
Number of leads	12 (I, II, III, aVR, aVL, aVF, V1-V6)
Diagnostic superclasses	5 (NORM, MI, STTC, CD, HYP)
Mean age (years)	59.6 (SD 16.9)
Mean height (cm)	167.5 (SD 7.8)
Mean weight (kg)	71.7 (SD 11.6)

Table 3: PTB-XL dataset summary statistics

4.2 Training Methodology

The hybrid CNN-SE-Transformer model was trained using the Adam optimizer with a learning rate schedule that combines warm-up and cosine decay. The focal loss function with $\gamma=2.0$ and $\alpha=0.25$ was employed to address the significant class imbalance in the PTB-XL dataset, with label smoothing ($\epsilon=0.1$) applied to improve generalization and prevent overconfident predictions. Data augmentation included random amplitude scaling and additive Gaussian noise to improve robustness. The signal preprocessing pipeline (bandpass filtering at 0.5-45 Hz, db4 wavelet denoising, and z-score normalization using training set statistics) was applied consistently during both training and inference. The model architecture was saved in the native Keras 3.x .keras format, with custom objects (focal loss, SE block, Transformer encoder) registered for proper deserialization.

4.3 Classification Performance

The model achieves robust multi-label classification performance across all five diagnostic categories. Per-class F1-optimized thresholds were determined through systematic evaluation on the validation set, balancing precision and recall for each class individually. The use of focal loss with label smoothing was critical for handling class imbalance, particularly for the underrepresented HYP and CD classes. The SE attention mechanism improved feature discrimination by adaptively weighting channel contributions, while the Transformer encoder enhanced the capture of long-range temporal patterns essential for detecting subtle morphological changes in conditions such as STTC. The binary pre-filter model achieves high sensitivity for the normal/abnormal distinction, ensuring that pathological cases are rarely missed in streaming mode. The following table summarizes the per-class classification thresholds and their relationship to the model confidence.

Class	Thresho ld	Optimization Criterion	Key Features
NOR M	0.61	High precision to minimize false negatives	Regular rhythm, normal P-QRS-T morphology
MI	0.49	Balanced precision-recall for clinical urgency	ST elevation, pathological Q waves, reciprocal changes
STTC	0.44	Higher recall for broad abnormality capture	ST depression, T-wave inversion, nonspecific changes
CD	0.35	Maximum recall for subtle conduction defects	Bundle branch blocks, AV blocks, axis deviation
HYP	0.42	Balanced threshold for moderate prevalence	Increased voltage criteria, axis deviation, repolarization changes

Table 4: Per-class threshold optimization criteria and key diagnostic features

4.4 XAI Visualization Results

The three XAI methods provide complementary insights into model predictions. Grad-CAM heatmaps effectively highlight the temporal regions within each lead that most strongly influence the predicted class. For MI predictions, the heatmaps consistently highlight the QRS complex and ST segments in inferior and lateral leads, aligning with established clinical knowledge of infarct localization. Saliency Maps provide a finer-grained attribution, revealing individual time points with high gradient magnitude. The Integrated Gradients analysis yields both lead-level and metadata-level importance scores. Lead importance analysis shows that leads V1-V3 and II are consistently among the most influential for MI detection, while leads I and aVR carry less diagnostic weight, consistent with the spatial sensitivity of these leads. Metadata importance analysis reveals that age and heart rate are the most influential demographic features, with sex and height/weight providing secondary contributions. These findings align with clinical understanding and provide interpretable evidence supporting model predictions.

4.5 LLM Report Quality Assessment

The fine-tuned GPT-4o-mini model generates clinically structured reports that integrate diagnostic predictions with patient context. Each report follows a four-section structure: Diagnostic Summary, Clinical Correlation, Possible Symptoms, and Recommendations. The system prompt enforces medical accuracy and professional tone, while the user prompt provides the necessary clinical context including predicted conditions, confidence scores, heart rate, rhythm, and patient demographics. The streaming interface delivers the report incrementally, with typical generation completing within 5-10 seconds. When multiple conditions are predicted, the report explains their potential clinical relationships, providing contextually rich interpretations. The fallback report generator ensures that a basic clinical summary is always available, even when the LLM API is unreachable, maintaining system reliability under adverse conditions.

4.6 Streaming Pipeline Evaluation

The streaming pipeline was evaluated on synthetic and real continuous ECG signals. The binary pre-filter effectively identifies normal windows with high confidence, enabling significant computational savings in monitoring scenarios where the majority of windows represent normal rhythm. Window-level analysis captures transient abnormalities that might be missed in single-record analysis, and the 50% overlap ensures that events occurring at window boundaries are not lost. The real-time visualization updates per-window with predictions, heart rate, rhythm assessment, and anomaly flags, providing clinicians with immediate feedback. The analysis history table enables retrospective review of all processed windows, and the downloadable stream report supports longitudinal documentation.

Chapter 5: Conclusions and Future Work

5.1 Summary of Achievements

CardioInsight demonstrates that the integration of deep learning classification, multi-method explainability, and LLM-powered reporting can produce a clinically useful and technically transparent ECG diagnosis system. The hybrid CNN-SE-Transformer model achieves multi-label classification across five diagnostic categories with per-class F1-optimized thresholds, addressing the inherent class imbalance in real-world ECG data through focal loss and label smoothing. The three complementary XAI methods (Grad-CAM, Saliency Maps, and Integrated Gradients) provide multi-level interpretability, from temporal region highlighting to per-lead and per-feature importance scores, bridging the gap between black-box predictions and clinical transparency. The fine-tuned GPT-4o-mini report generator synthesizes structured clinical interpretations that integrate predictions with patient context, producing reports that mirror professional cardiology documentation. The streaming pipeline with binary pre-filtering enables real-time monitoring with computational efficiency, supporting the deployment of continuous cardiac surveillance in clinical and remote settings.

5.2 Limitations

Several limitations should be acknowledged. The model was trained exclusively on the PTB-XL dataset, which, while being the largest publicly available ECG dataset, may not fully represent the diversity of cardiac conditions encountered in clinical practice, particularly rare arrhythmias and pediatric ECGs. The system is designed as a clinical decision support tool and must not be used as a substitute for professional medical diagnosis. The LLM-generated reports, while structurally consistent with professional standards, may occasionally contain clinically inaccurate assertions and should be reviewed by qualified physicians. The streaming pipeline currently operates on pre-recorded signals loaded from .npy files, and integration with live hardware acquisition would require additional engineering. The binary pre-filter adds latency to normal window classification, and its accuracy directly impacts the sensitivity of the streaming system for detecting subtle abnormalities.

5.3 Future Work

Several directions for future development have been identified. First, integration with real-time hardware ECG acquisition devices would enable true continuous monitoring in clinical environments, requiring the development of a device driver layer and real-time signal buffering. Second, extending the classification taxonomy beyond the five PTB-XL superclasses to include sub-diagnoses (e.g., anterior vs. inferior MI, left vs. right bundle branch block) would increase clinical utility. Third, federated learning approaches could enable model improvement across

institutions without sharing sensitive patient data, addressing privacy concerns while expanding the training data distribution. Fourth, incorporating additional modalities such as echocardiogram features or cardiac biomarkers could improve diagnostic accuracy through multimodal fusion. Fifth, prospective clinical validation studies comparing CardioInsight predictions with expert cardiologist interpretations would provide the evidence base necessary for regulatory approval and clinical deployment. Finally, localization of the interface and reports to Arabic and other languages would enhance accessibility in the MENA region, aligning with the AIO mission of promoting AI solutions for the Arab world.

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